Workshop tools & machines

National 5 · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name and describe the use of saws, chisels, planes, hammers and mallets
- ☐ name and describe the use of drill bits: twist, Forstner, countersink, centre
- ☐ set up and use the pillar drill safely, including the depth stop
- ☐ describe centre lathe setting up, parallel, step and taper turning, drilling and knurling
- ☐ describe wood lathe setting up, preparing the blank, parallel turning, parting off and finishing
- ☐ describe the use of taps and dies, the strip heater, oven and fluidiser
- ☐ name the correct holding tool: vices, G-clamp, sash cramps, formers and jigs

I DO — Worked example

Read how a full-mark answer is built

A rectangular picture frame has been glued together.

1 Describe two methods of checking that the frame is square. (2 marks)

Full-mark answer each ✓ is one mark

Measure across both diagonals with a steel rule; if the measurements are equal the frame is square ✓.
Place a try square into each inside corner and check there is no gap against the blade ✓.

Why marks are not awarded

"Use a ruler to measure it and check it."

0 marks: does not say what is measured or how it shows squareness.

WE DO — Guided practice

Use the starters and word bank to help you

Holes for a set of pegs are drilled 15 mm deep into a beech rail on the pillar drill.

2 Outline how the holes could be drilled to the correct depth. (1 mark)

Sentence starters

One … is …

Word bank

Depth stop · required · masking · pillar drill

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which saw is best for cutting a curve in 6 mm acrylic?
☐ Tenon saw ☐ Coping saw ☐ Hacksaw ☐ Panel saw

b) How is a hole of a set depth drilled consistently on a pillar drill?
☐ Use a centre punch ☐ Set the depth stop ☐ Use a larger bit ☐ Use soft jaws

c) Which centre lathe process creates a diamond grip pattern?
☐ Parting off ☐ Knurling ☐ Facing ☐ Taper turning

d) Which tool would you use before drilling a hole in mild steel?
☐ Bradawl ☐ Centre punch ☐ Marking knife ☐ Mortise gauge

4 Key terms: write the letter of the matching description next to each term.

1. **Marking gauge**	____	**A** Drills clean, flat-bottomed holes in wood.
2. **Mortise gauge**	____	**B** Cutting curves in thin wood or plastic.
3. **Odd-leg callipers**	____	**C** Marks two parallel lines for a mortise and tenon (two spurs).
4. **Centre punch**	____	**D** Scribe a line parallel to the edge of metal.
5. **Coping saw**	____	**E** Marks a line parallel to an edge on wood (one spur).
6. **Forstner bit**	____	**F** Makes a dimple to locate a drill before drilling metal.

Before turning a mild steel handle on the centre lathe the pupil carries out safety checks.

5 State two safety checks that should be made on the centre lathe before turning. (2 marks)

The aluminium bracket now has to be bent to 90°.

6 Describe how the bracket would be bent. (2 marks)

An M6 internal thread is cut into the base of a lamp.

7 Describe two ways of making sure a good quality internal thread is produced. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?